Systems Analysis and Design: Bus System for the University of North Alabama

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Section I: Problem Statement and Feasibility Study

The University of North Alabama was founded in 1830 and called LaGrange College. Over its long history it has undergone numerous changes and expansions, especially in student population and campus size. As of 2024, the University of North Alabama had over 9,000 students. In 2011, the University of North Alabama (UNA)began implementing a bus system available to students, with routes and stops centered around the apartment complexes neighboring the university. At the time, the university offered a night bus service to students, which garnered success. They expanded the route to cover trips to East Campus and the McKinney Department of Human Sciences. The university had a program called 'Lion's Lift,' a shuttle service operating around campus that could be booked online; however, this system is no longer in service as of August 2023. Currently, there is minimal transportation available across the campus. The only route information available is a PDF consisting of three rarely updated stops, making it difficult for users of the transportation system to utilize the system effectively. Additionally, it is difficult for students and faculty to know where and when they can be picked up and dropped off.

Creating a project of this nature will prove to be economically feasible. Considering the current economic climate, the percentage of students living on or around campus without a car is likely to increase. Therefore, the necessity of public transportation and a way to access said transportation is clear. The implementation of this system will significantly involve the cooperation of UNA faculty , considering that the university currently operates the transportation system. Important transportation routing information can also be sourced from the university. The cost of collecting the necessary staff and infrastructure is approximately $50,000 to $100,000 in U.S. dollars. The projected increase in ridership of UNA shuttles nets the ability to offset this cost with advertisement. Implementing this new system with current transportation options is feasible. Our system does not interfere with the current shuttle system. The drivers of these vehicles are likely already monitored for shift purposes. This product will simply gather live location information to provide feedback to students and staff. The proposed system may provide information to optimize routes, and drivers a feature especially useful with the dynamic roadways surrounding UNA. Continuing a particular service has proven unreliable concerning the overall trend of the University of North Alabama's public transportation. The ‘Slidr’ service (a temporary contract that functioned similarly to “Uber”) ended abruptly when the contract was not renewed. Despite the cancellation of “Slidr” shuttle services have been steadily operating around campus for over a decade. The estimated time to completion of this project will not hinder the overall outcome since the project is a high priority. The project team has also proven effective in current tasks, significantly affecting the operation's success. With proper collaboration between the university and the city of Florence, the operation should proceed smoothly. This project will require approval from the University of North Alabama to gather and store their transportation information properly.

The systems and agreements currently in place for the university’s transportation must be intact. The system requires the university’s approval and to host the application on their website. The time needed to obtain this approval is well within time constraints. Obtaining these checkpoints is also built into the budget. These will ensure proper function and legality in our creation. A project of this nature requires between 4 and 6 months to develop a production-level system. This timeframe aligns with the course's constraints. The readiness of the staff and outside sources will determine precisely how quickly this system can be built. The system may produce results faster, depending on the quality of its infrastructure and testing environment, which will be adequately sourced according to the budget.

We aim to introduce several new concepts to provide students and faculty with reliable transportation. This system will provide live updates on the buses around campus, utilizing a Global Positioning System (GPS) hosted on the Internet. This feature allows users to know when they will be picked up, where the buses on the route are located, and how far away each bus is. The GPS feature allows the system to provide constant feedback to bus app users. Additionally, the live updates will inform users if there has been a delay in transportation, whether due to a system complication or a road delay. There will be more stops than three, unlike the old transportation system. Overall, the app used for this system will provide users with the necessary information for the bus system, including wait times, stops, arrivals, and even the ability to create their own stops.

Section II: Brief Review of Requirements Analysis and Specification

We do have requirements that we want to ensure are integrated into the bus's GPS subsystem design. The subsystem related to GPS will enable wait times to be created in the app for students and faculty. The location update will be performed using the location API every sixty seconds. The Location API is a service that enables users, specifically students and faculty, to determine their location. The GPS will not collect any data between 17:00 and 6:00 Central Standard Time (CST) on weekdays. The GPS will turn on at 6:00 CST and off at 17:00 CST. The bus system is not in operation on weekends; therefore, no GPS data is collected during this time.

This bus system will integrate with a backend system. A backend system comprises all the components, such as setup or code, that enable the application to run, but these components are not accessible to users, including students and faculty.

Additionally, there are requirements that we will aim to meet when integrating the bus's GPS subsystem. This subsystem should not cost more than $35,000 U.S. dollars to develop and should not exceed $5,000 U.S. dollars in annual operating costs. The GPS subsystem will have a yearly uptime of 95%, which allows for approximately 18.25 days of downtime. The components within the subsystem will automatically check daily for any necessary system updates and download and install any required software updates.

The GPS subsystem is a web-based system with local installation on the physical buses on the campus. Each day, the shuttle driver starts the GPS when the bus begins for the day. The University of North Alabama's Information Technology (IT) department will maintain the systems. Other systems will interface with the GPS subsystem, such as GPS transmitter hardware, a localized network on the shuttle, online-hosted GPS, and an end user's device where it is displayed. The GPS transmitter hardware features an antenna that receives the signal and a processor that converts the signal into valuable data. The data is then displayed on a display unit, which in this case is likely to be a smartphone or computer, for the user. While the GPS subsystem updates the shuttles, the regular route updates are provided by the University of North Alabama transportation services.

Section III: Preliminary Design

For the research, the University of North Alabama's bus system application has at least one subsystem. This subsystem goes into the global positioning system (GPS) for the bus app. The estimated arrival times are displayed to users via the Internet, based on GPS data. These times pulled for the GPS will be updated every minute to ensure up-to-date and realistic wait and arrival times. The application may be written as an app for both iOS and Android operating systems; therefore, the code will be written twice. This app allows users to request information from the back end, such as finding estimated bus arrivals, viewing live bus locations, and displaying any issues that may occur with the bus's travel, including accidents or reroutes. Users will be able to interact and look up the future wait times and stops for the bus(es).

Several things are involved in this system, including the students and faculty, the GPS receiver program on the internet, and the end user's device(s). The system features a local installation within the buses, including the GPS transmitter. There is a subsystem, the GPS, for the bus system. The GPS is a web-based system. Every morning, this process begins when the bus drivers start their buses. Some systems that interface with the system include the GPS transmitter hardware, a localized network on the bus, a GPS subsystem, and the end user's device, where the app is displayed. A GPS transmitter is a device that determines and sends back the location and data using the Global Positioning System (GPS). A localized network, also known as a local area network (LAN), is a collection of devices that are interconnected within a specific area. The end user's device, such as a smartphone, is the device used by a student or faculty member to view the bus app. Therefore, the end user is the one who extracts the GPS information as the buses go through their routes. The bus system will pull data live from the GPS to update the maps and bus locations in the app. The UNA transportation service administrators update the bus routes, which can be done in the bus app. Additionally, the app features are tailored to users, including students and faculty. However, these features are also available to service administrators and admins.

Section IV: Detailed Design

Relevant stakeholders in the University of North Alabama bus system are students and faculty, UNA's IT department, and the bus driver(s). However, stakeholders' expectations differ. Students and faculty expect to have a reliable and efficient way of transportation around campus. The time between two classes is typically ten to fifteen minutes. The bus system can be beneficial if one's buildings are located across the campus. Additionally, this system may be helpful to those unfamiliar with the university's layout, such as visitors or students new to the Florence area. The expectations of the University of North Alabama's IT team are that they should be able to complete the required maintenance of the bus system, such as working with the GPS hardware and the system's applications. Maintenance jobs may require ensuring that the web-based GPS is functioning correctly. This includes the system connected to the Internet, and times are updated accordingly. Additionally, the system has transmitter hardware and a LAN network on the buses, which must also be maintained. Nonetheless, problems may arise in the system that are known. These issues include the GPS going down, buses being inaccessible, or insufficient funds. However, these give the system room to grow and improve for the university's sake.

Several components are involved in this bus system. The key elements are the users' browsers, UI, and business logic components, centralized shuttle databases, and GPS data. UI stands for user interface. The user interface involves the user interacting with the device or software. This may include buttons, login menus, navigational bars, and other elements. Business logic deals with the software knowing when and how to perform specific tasks within the app, such as creating a stop. In the admin menu, this includes managing, adding, or deleting stops for the bus schedules. A centralized database is a system where all the data is stored in one location, typically on a server. The data from the GPS allows the app to read the location of the buses. Additionally, some components can be bought. These components include the GPS data provider, map and route services, an authentication system, and application hosting. The authentication system would help the system verify the user logging in. The login credentials for this system are the user's UNA username and password. Application hosting is a service that allows us to host and run software applications. All of these components, together, would make an efficient and effective bus system for the University of North Alabama.

Section V: Project Management

Several charts were created to ensure that this system would be beneficial in the future and to determine how long it would take for the costs to break even. These were done with a net present value table, a payback period table, and a return on investment table. Each of these tables serves a purpose in enhancing the understanding of the project's financial standpoint. To start, a net present value (NPV) table was created. The purpose of the net present value table is to help a team or group visualize whether the project will be profitable over time. In this system, the initial investment is $90,000 and lasts for the first six years. Additionally, over these years, the initial investment is accompanied by an eight percent discount rate. With this rate, the initial investment has recovered between years two and three, at which point the cumulative NPV turns positive. At year six, the total present value of the asset is $461,565, which proves to be highly beneficial in the future.

Secondly, a payback period table helps visualize how quickly a project will recover from the initial costs. Similarly to the NPV table, the payback period is between the second and third year. This is where the cumulative total turns positive, going from -$62,220 to $69,690. With these, a return on investment table was created for this system. A return on investment table helps visualize the profit from the system. Furthermore, this can help monitor the performance of a system, indicating whether it is running successfully or not. The total amount of net pay that is discounted is $325,236, which is $222,282 at the 8% discount. Therefore, the undiscounted amount yields a 360% return, which is equivalent to a 247% return at an 8% discount.

With this system, a PERT chart and a CPM chart were created. These two charts are designed to help plan the estimated time required for each task to fit within the project's time window requirement. As mentioned, this system is expected to be implemented within six months. The main difference between these two systems is that the CPM chart has a fixed time estimate versus a calculated estimate based on three time estimates: optimistic, pessimistic, and most likely. For this system, the critical path is estimated to be around seventeen weeks. The critical path for these charts is the path that takes the longest time, which indicates the shortest time estimate for project completion. Additionally, any delays in these tasks will set back the entire project. For this system, the critical path includes five steps. These steps are system design, backend development, system integration, system testing, and system deployment. There are two additional paths for the system, which are not part of the critical path. However, all of the paths have the same starting point, which is system design. One of the additional paths includes system design, app development, system integration, system testing, and system deployment. The second additional path provides system design, acquiring the buses, and hardware setup. The hardware setup phase splits into two paths: one for system testing and the other for bus driver training. Both of these lead to system deployment. Similarly to the beginning, all of the paths end with system deployment.

Section VI: Mockup of Prototype

The mockup of the prototype for the bus system indicates what the bus system app would look like. There is currently no effective transportation system for students and staff to travel across the campus. For our prototype, there are differing ways the app can look, depending on who is using it: the admin or the users, who are students and faculty. Either way, the first step is to log in using your UNA credentials. First, this discussion will focus on the administrator's perspective on the shuttle bus app. Once the admin has logged in, they are presented with a live map of the campus, accompanied by a navigation bar in the top right corner. Once the navigation bar is selected, the admin(s) can select 'View All Buses.' This selection will direct them to a list of buses used for the campus, along with the estimated time of arrival at their next stop. From here, the admin can edit the bus, such as its bus number and start and destination points, or they can delete the bus. Whenever a bus is deleted, it removes any trace of the bus from the app; therefore, no confusion should occur for students or faculty. Deleting a bus would be an occurrence if a bus were to be out of commission or no longer needed. A second option for the admin to select from the navigation bar on the home screen is 'View All Stops.' As with the buses, this provides the admin(s) with a list of all bus stops, allowing them to edit or delete stops as needed. The students' views of those options are similar; however, they do not have the option to edit or delete the buses or stops.

Additionally, there is an option called 'Plan A Stop.' In the 'Plan A Stop' screen, the user gets two drop-down arrows for pick-up and drop-off locations. When a dropdown option is selected, the user will get a list of the available options. After setting up their desired stop, they hit a ‘Submit” button to confirm it. After ‘Submit’ has been selected, the user gets an overview of the stop they planned. This includes their pick-up location, drop-off location, and which bus will be arriving in a few minutes. Furthermore, the ‘Stop Guidance’ option is listed below the information. When this button is selected, it will take the user back to the map screen with the navigation bar.